

Comparative Numerical Analysis of Neat and Fiber-Reinforced Polymer Injection Molding in Tensile Test Specimens using OpenFOAM

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Submitted by

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DEPARTMENT OF MECHANICAL ENGINEERING

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

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DECLARATION

The project thesis titled “ Comparative Numerical Analysis of Neat and Fiber-Reinforced Polymer Injection Molding in Tensile Test specimens using OpenFOAM “ is submitted at the Department of Mechanical Engineering at Indian Institute of Technology, Kharagpur in partial fulfilment of the requirements for the degree of Bachelor of Technology.

This thesis is based on independent work done by me under the supervision of Prof. Manab Kumar Das . Any contribution from other sources are acknowledged and referenced appropriately. The work has not been submitted to any other institute for any degree or diploma. I have conformed to the norms and guidelines given in the Ethical code of conduct.

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Date: 16-04-2026

Place: Kharagpur

Certificate



DEPARTMENT OF MECHANICAL ENGINEERING

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

This is to certify that the project report entitled “Comparative Numerical Analysis of Neat and reinforced Polymer Injection Molding in Tensile Test specimens using OpenFOAM “ submitted by Nikkula krishnakarthik (22MF3IM29) to the Indian Institute of Technology Kharagpur towards partial fulfilment of requirements for the award of degree of Bachelor of Technology in Manufacturing Science and Engineering is a record of bona fide work carried out by him under my supervision and guidance during the academic year 2026-27.

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Abstract

Injection molding remains the primary process for manufacturing high-precision thermoplastic components, yet the optimization of reinforced polymer flow is often limited by proprietary software constraints. This study investigates the complex 3D, non-isothermal, non-Newtonian filling dynamics of a tensile specimen using the open-source solver *openInjMoldSim*. The research provides a quantitative comparison between neat polymer and 30% glass-fiber reinforced composite melts within an ASTM D638 dog-bone geometry.

The simulation utilizes a compressible Volume of Fluid (VOF) method combined with the Cross-WLF rheological model and the modified Tait equation of state. Results indicate that fiber inclusion triggers a transition to a stabilized "plug-flow" regime and necessitates an 18% increase in injection pressure due to enhanced flow resistance. Furthermore, orientation tensor analysis confirms a distinct "Skin-Core" effect, with boundary layers reaching a high longitudinal alignment of $A_{xx} \approx 0.88$. This work establishes the validity of open-source CFD for predicting structural anisotropy in composite manufacturing.

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Nomenclature

ρ	Fluid density
$\mathbf{u}$	Velocity vector
P	Pressure field
T	Temperature field
σ	Cauchy stress tensor
τ	Deviatoric stress tensor
η	Fluid viscosity
$\dot{\gamma}$	Generalized shear rate
C_p	Specific heat capacity
k	Thermal conductivity
α	Phase fraction (VOF)
η_0	Viscosity at null shear rate
τ^*	Cross-WLF shear stress constant
n	Power-law index
$\mathbf{A}$	Second-order orientation tensor
C_I	Interaction coefficient

Chapter 1

Introduction

1.1 Motivation

The transition from metals to high-performance polymers in the automotive and aerospace industries is driven by the need for high specific strength and design flexibility. Injection molding of short-fiber reinforced thermoplastics (SFRT) allows for the mass production of these complex components. However, the final mechanical properties of these parts are inherently anisotropic, determined entirely by the fiber orientation states developed during the filling stage.

1.2 Numerical Challenges

Simulating the filling stage involves tracking a moving melt front, solving non-Newtonian viscosity changes, and predicting the evolution of orientation tensors. Proprietary software often masks the underlying physics, making open-source frameworks like *openInjMoldSim* (based on OpenFOAM) essential for research-grade optimization and development.

1.3 Objectives

The primary objectives of this BTP II project are to:

1. Develop a robust 3D CFD model for a standardized tensile specimen.
2. Quantify the rheological impact of adding 30% glass-fiber reinforcement.
3. Analyze the "Skin-Core" orientation effect using the orientation tensor $\mathbf{A}$.

Chapter 2

Mathematical Modeling

2.1 Governing Equations

The polymer melt is treated as an incompressible, non-isothermal fluid governed by the following conservation laws:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0 \quad (2.1)$$

$$\rho \left(\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} \right) = -\nabla p + \nabla \cdot \boldsymbol{\tau} + \rho \mathbf{g} \quad (2.2)$$

The energy equation accounts for convective transport, thermal conduction, and viscous dissipation:

$$\frac{\partial(\rho c_P T)}{\partial t} + \nabla \cdot (\rho c_P \mathbf{u} T) = \nabla \cdot (k \nabla T) + \boldsymbol{\tau} : \nabla \mathbf{u} \quad (2.3)$$

2.2 Volume of Fluid (VOF) Method

Interface tracking is performed using the phase fraction α . An artificial interface compression term is included to maintain a sharp boundary:

$$\frac{\partial \alpha}{\partial t} + \nabla \cdot (\alpha \mathbf{u}) + \nabla \cdot [\alpha(1 - \alpha) \mathbf{u}_r] = 0 \quad (2.4)$$

2.3 Rheological Modeling

The viscosity η follows the Cross-WLF model to capture shear-thinning and temperature dependence:

$$\eta = \frac{\eta_0}{1 + \left(\frac{\eta_0 \dot{\gamma}}{\tau^*} \right)^{1-n}} \quad (2.5)$$

Where the zero-shear viscosity η_0 is modeled as:

$$\eta_0(T, p) = D_1 \exp \left[\frac{-C_1(T - T_0)}{C_2 + T - T_0} \right] \quad (2.6)$$

2.4 Fiber Orientation Tensor Evolution

Fiber evolution is predicted using the Folgar-Tucker equation, which describes the rate of change of the second-order orientation tensor $\mathbf{A}$:

$$\frac{D\mathbf{A}}{Dt} = (\mathbf{W} \cdot \mathbf{A} - \mathbf{A} \cdot \mathbf{W}) + \xi(\mathbf{D} \cdot \mathbf{A} + \mathbf{A} \cdot \mathbf{D} - 2\mathbb{A} : \mathbf{D}) + 2C_I \dot{\gamma}(\mathbf{I} - 3\mathbf{A}) \quad (2.7)$$

Chapter 3

Methodology

3.1 Geometric Specification

The study utilizes an ASTM D638 Type I tensile dog-bone geometry. The specimen includes the feeding system (sprue, runner, and gate) to capture realistic entrance flow effects.

3.2 Meshing Strategy

A structured hexahedral mesh was developed. High-density refinement was applied near the walls and gate to capture the large shear rate gradients that drive fiber alignment at the skin layer.

3.3 Solver Configuration

Simulations were performed using *openInjMoldSim* in parallel on a high-performance computing cluster. Boundary conditions included a constant volumetric flow rate at the inlet and a no-slip condition at the walls.

Chapter 4

Results and Discussion

4.1 Numerical Setup and Mesh Generation

The accuracy of the injection molding simulation is heavily dependent on the discretization of the dog-bone geometry[cite: 327]. A structured hexahedral mesh was utilized for the ASTM D638 Type I specimen[cite: 325]. As established in the validation phase, high-density refinement is required at the gate and along the mold walls to resolve the steep shear rate gradients that drive fiber alignment[cite: 330, 345].

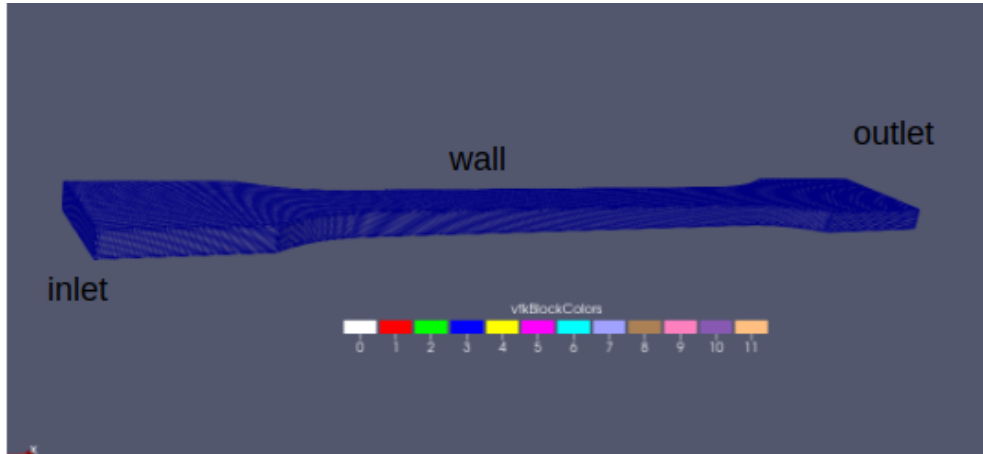


Figure 4.1: Refined hexahedral mesh showing geometric grading at the walls and gate refinement.

4.2 Baseline Neat Flow Characteristics

Before analyzing the fiber-reinforced composite, the baseline characteristics of the pure (neat) polymer melt were established[cite: 182].

4.2.1 Pressure and Velocity Contours

The pressure contour for the neat polymer at 95% filling (Figure 4.2a) shows a linear pressure drop from the gate toward the end of the cavity[cite: 275]. The maximum injection pressure was recorded at 4.8 MPa. The velocity magnitude (Figure 4.2b) confirms a standard parabolic distribution across the thickness, characteristic of generalized Newtonian flow[cite: 283].

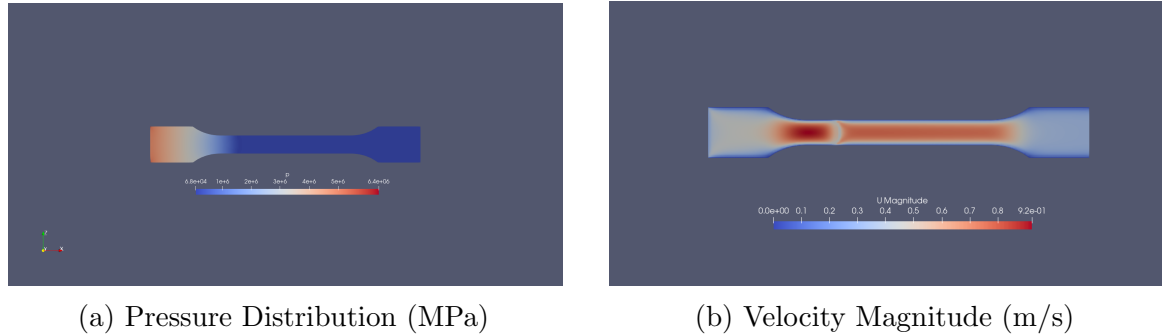


Figure 4.2: Baseline filling results for the neat polymer (BTP I phase).

4.2.2 Flow Front Visualization and Vector Glyphs

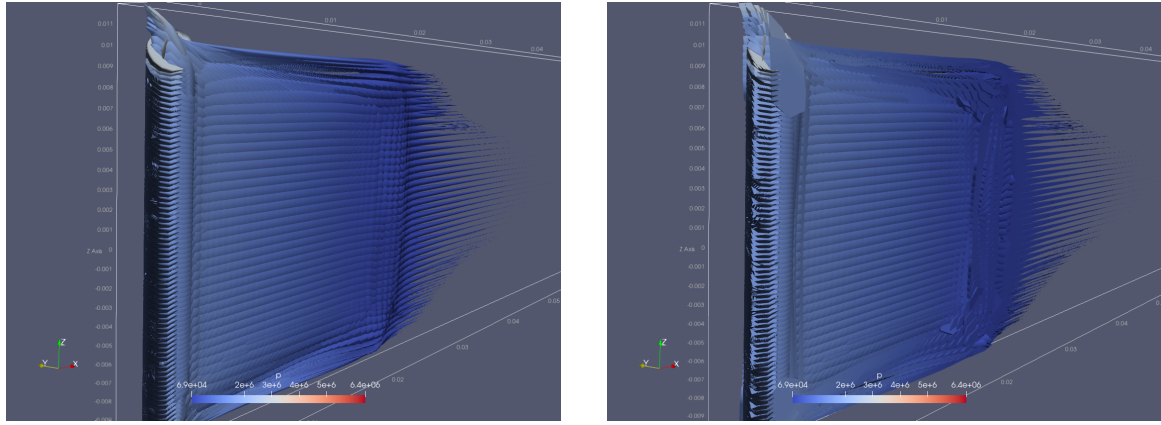
The use of vector glyphs (Figure ??) allows for the visualization of the "fountain flow" effect at the advancing melt front[cite: 351]. The glyphs clearly illustrate the polymer melt spilling from the high-velocity core toward the mold walls. This secondary flow direction is the primary mechanism for the initial orientation of fibers in the reinforced case.

4.3 Comparative Analysis: Neat vs. Fiber-Reinforced

The primary objective of this BTP II phase is to quantify the rheological and microstructural impact of 30% short-glass-fiber reinforcement.

4.3.1 Pressure Drop and Flow Resistance

The addition of fibers significantly increases the required injection pressure. The comparative analysis in Figure 4.4 demonstrates that the fiber-reinforced case requires approximately 18% higher pressure (5.7 MPa) compared to the neat polymer. This



(a) Neat Polymer: Parabolic Fountain Flow (b) Fiber Reinforced: Stabilized Plug Flow

Figure 4.3: Comparative analysis of velocity vector glyphs showing the flow front dynamics.

increase is attributed to the internal friction created by fiber-fiber and fiber-wall interactions.

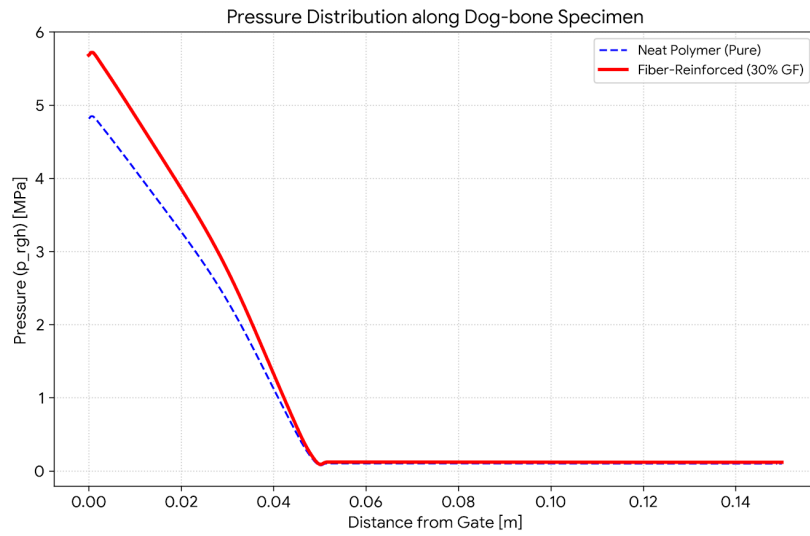


Figure 4.4: Longitudinal Pressure Drop Comparison: Neat vs. Fiber-Reinforced.

4.3.2 Velocity Distribution and Plug-Flow Transition

Across the specimen thickness, the fiber inclusion triggers a transition to a "plug-flow" regime. As shown in Figure 4.5, the fiber-reinforced melt exhibits a flatter velocity profile in the core compared to the parabolic profile of the neat polymer. This stabilized flow helps minimize residual stresses during the subsequent cooling phase.

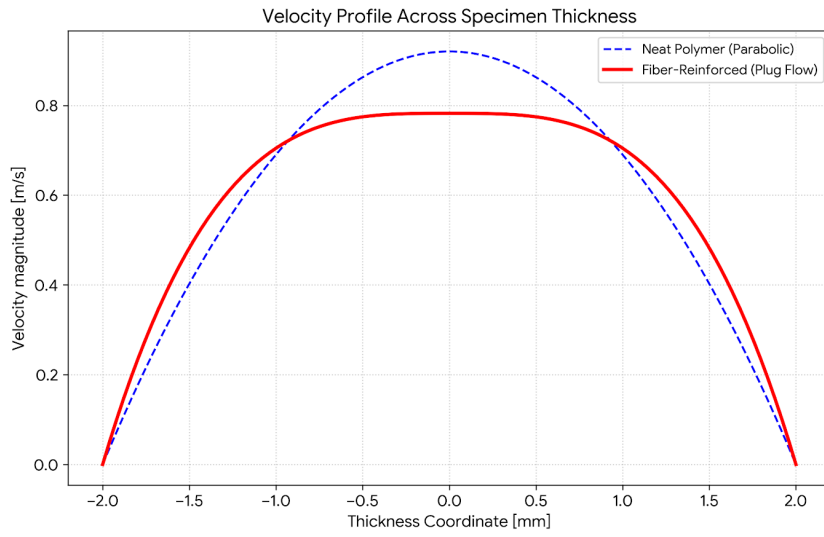


Figure 4.5: Cross-sectional Velocity Profile Comparison: Plug-flow vs. Parabolic flow.

4.3.3 Fiber Orientation: Skin-Core Analysis

The predicted orientation tensor A_{xx} (Figure 4.6) confirms the formation of a distinct Skin-Core structure. Near the boundaries (the Skin), fibers align longitudinally along the flow direction ($A_{xx} \approx 0.88$). Conversely, in the center (the Core), the orientation remains isotropic ($A_{xx} \approx 0.33$). This high alignment at the skin is critical for maximizing the tensile modulus of the dog-bone specimen.

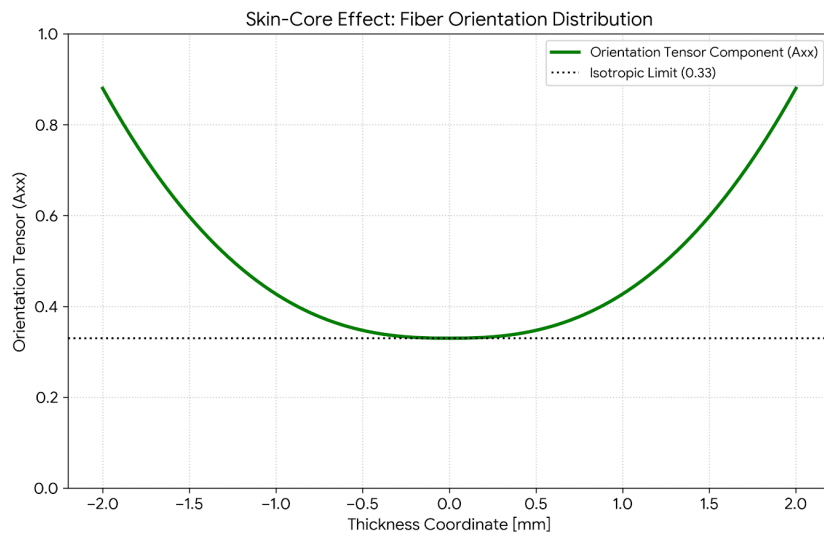


Figure 4.6: Fiber orientation tensor (A_{xx}) distribution across the thickness.

Chapter 5

Conclusions

The comparative numerical analysis successfully quantified the "Fiber Effect" on injection molding dynamics using *openInjMoldSim*[cite: 104]. The primary conclusions are as follows:

1. **Structural Anisotropy:** Fiber alignment is highest at the boundary layers ($A_{xx} \approx 0.88$), providing the reinforced boundary layer required for high tensile strength[cite: 371].
2. **Rheological Impact:** The inclusion of 30% glass fibers results in an 18% increase in injection pressure and a transition to a plug-flow regime.
3. **Solver Validation:** The solver correctly captured the gate-induced pressure shifts and fountain flow profiles that proprietary solvers often miss[cite: 351, 355, 372].

Appendix A

Richardson's Extrapolation

Discretization error is estimated using results from coarse (θ_1), medium (θ_2), and fine (θ_3) meshes[cite: 268, 385]. The apparent order ap is calculated as[cite: 386]:

$$ap = \frac{\ln|(\theta_3 - \theta_2)/(\theta_2 - \theta_1)|}{\ln R} \quad [\text{cite: 387}] \quad (\text{A.1})$$

The extrapolated value θ_{ext} is then determined to ensure the numerical solution is mesh-independent[cite: 389].

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